E2: Neutrally Stratified Boundary Layer

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Introduction

- Neutral stratification : $\frac{\partial \theta}{\partial z} = 0$ or $\frac{\partial T}{\partial z} = -\frac{g}{c_p} \approx \frac{1\,K}{100\,m}$
- Flows are or can often be assumed to be stratified neutrally.
- Atmosphere: strict neutral stability conditions are rare. However, the ABL may be considered near-neutral especially during overcast skies and strong geostrophic winds.

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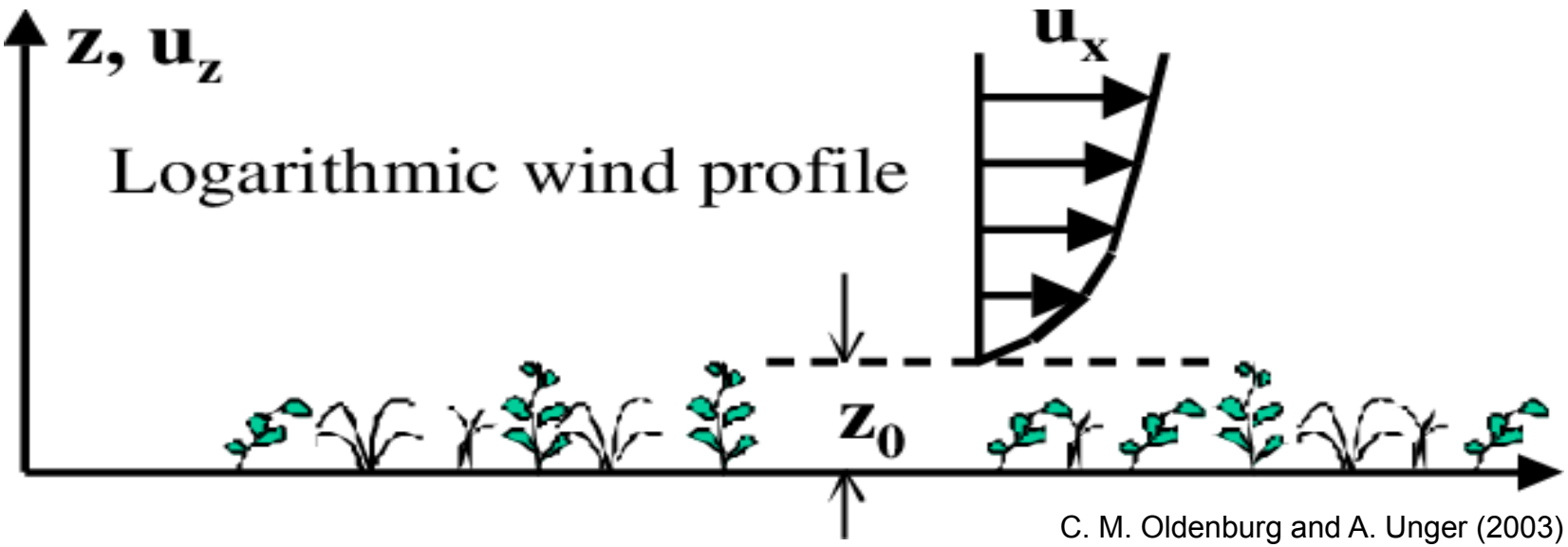
Introduction



C. M. Oldenburg and A. Unger (2003)

- Simpler theoretical and semi-empirical approaches exist for neutral boundary layers (e.g., logarithmic wind profile in the surface layer).
- Turbulence is entirely of mechanical origin and depends on the surface friction and vertical distribution of wind shear.

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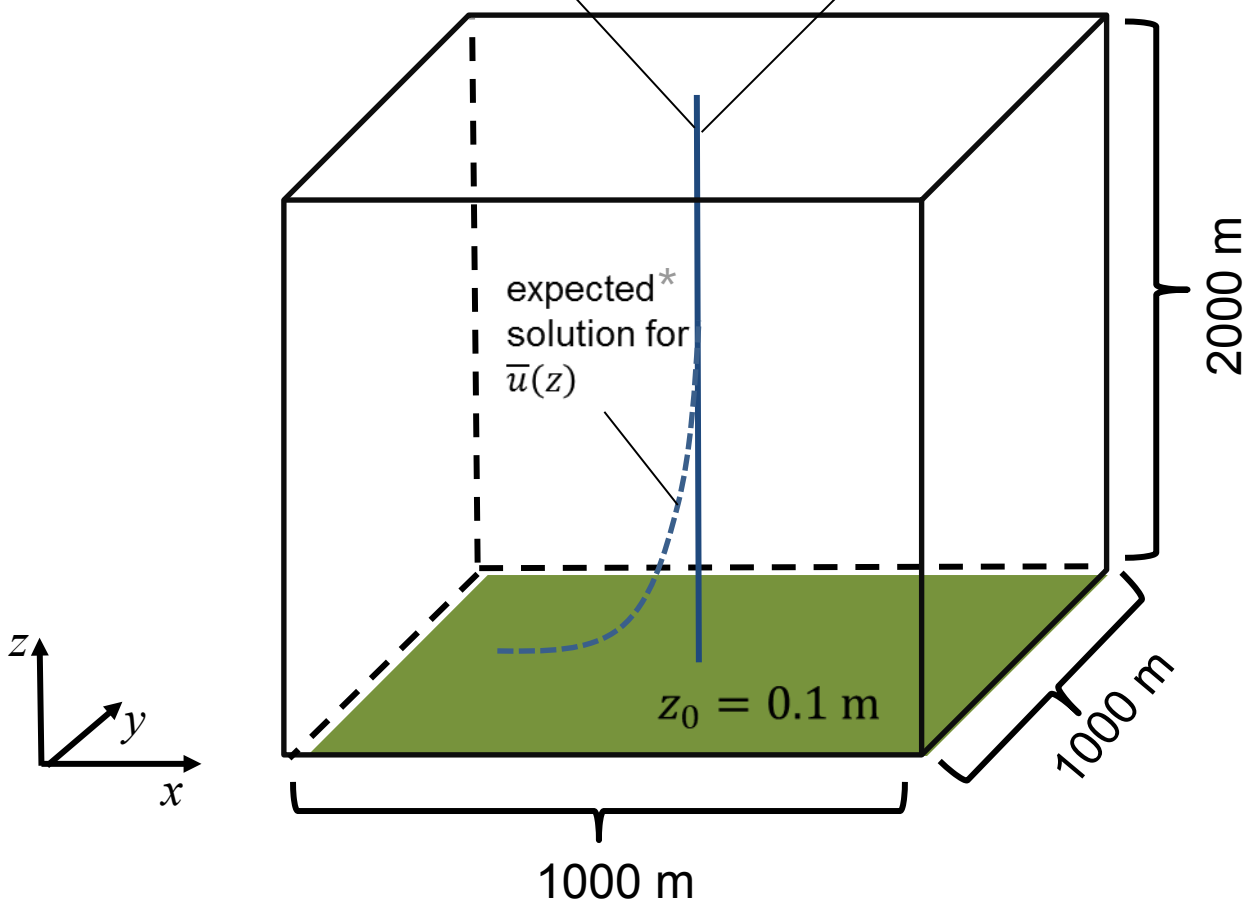
Task

Setup:

Initial temperature profile:
$\theta(z) = const = 300\,\mathrm{K}$

Initial wind profile:
$u(z) = const = u_g = 5\,\frac{\mathrm{m}}{\mathrm{s}}, \quad v(z) = v_g = 0$



- **Stratification:**
  Atmosphere is considered as neutrally stratified
- **Forcing:**
  Flow is driven by pressure gradient (geostrophic wind)
- **Grid size:**
  20m in all directions
- **Horizontal boundary conditions:**
  cyclic
- **Simulation time:**
  2 days

*: Please note, due to Coriolis force a wind turning with height will be expected resulting in a more complex profile for $v(z)$!

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Questions to be answered:

- After which time turbulence / mean flow becomes stationary?
- How do the horizontally and temporally averaged velocities and momentum flux profiles look like?
- Is it really a large-eddy simulation, i.e., are the subgrid-scale fluxes much smaller than the resolved-scale fluxes (except close to the surface)?
- How do the turbulence spectra of *u*, *v*, *w* along *x* and *y* look like?

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Hints (I)

- **Initial profiles/geostrophic wind**
  - Use `ug_surface`, `vg_surface`, `pt_surface` to prescribe the geostrophic wind & temperature.
  - With `initializing_actions = 'set_constant_profiles'` the initial wind profile will be constant with height with the geostrophic value.
- **Boundary conditions**
  - Between the bottom and the first grid level a constant flux layer (also called surface layer) shall be switched on, which corresponds to a no-slip condition at the surface. At the top, a Dirichlet condition with the geostrophic value shall be used. See `bc_uv_b` and `bc_uv_t` and `constant_flux_layer`.
- **Stationary state**
  - At the beginning of the simulation the mean wind in the boundary layer is not balanced with friction, Coriolis and pressure force → an inertial oscillation will appear. Due to friction the amplitude of the inertial oscillation becomes smaller with time till a stationary state is reached, but it would need very long time!
  - To prevent unnecessary long simulation time, a so-called Rayleigh damping can be applied, which damps the inertial oscillation. See parameters `rayleigh_damping_height` and `rayleigh_damping_factor`.

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Hints (II)

- **Saving CPU-time with Galilei transformation**
  - Switch on parameter `galilei_transformation` to save computing time.
  - Switch off the temperature equation to save CPU time by setting initialization parameter `neutral = .TRUE.`
- **Initial disturbances**
  - To initiate turbulence, it is necessary to impose artificial random perturbations to the horizontal velocity field (`create_disturbances, dt_disturb, disturbance_energy_limit`)
- **Spectra**
  - Spectra output is controlled by parameters `data_output_sp`, `dt_dosp`, etc. These package-parameters have to be given in a separate NAMELIST block:

```
&runtime_parameters end_time = ... , /
&spectra_parameters data_output_sp = ... , /
```

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Hints (III)

- **Palmrun command**
  - `palmrun  -r e2_neutral -a "d3#" -X4`
- **Palmplot commands**
  - `palmplot pr file_1=e2_neutral_pr.000.nc format_out=pdf file_out=e2_pr time_stride=2 no_rows=2 no_columns=3 var='u v w wu wv w*2'`
  - `palmplot pr file_1=e2_neutral_pr.000.nc format_out=pdf file_out=e2_pr_2 time_stride=2 var='w"u" w"v" w*u* w*v*'`
  - `palmplot ts file_1=e2_neutral_ts.000.nc format_out=pdf file_out=e2_ts var='E E* umax' no_rows=3`
  - `palmplot sp file_1=e2_neutral_sp.000.nc format_out=pdf file_out=e2_sp start_time_step=48 no_rows=2 no_columns=2`